

WASP-135b

R-0157

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_250209 · created 2026-07-16T15:04:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460716.008 +/- 0.013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.05
Transit depth (Rp/Rs)^2	2.49 +/- 1.59 [%]
Orbital Inclination (inc)	80.76 +/- 2.92
Airmass coefficient 1 (a1)	1.24 +/- 0.52
Airmass coefficient 2 (a2)	-0.229 +/- 0.068
Scatter in the residuals of the lightcurve fit is	35.5 %
Best Comparison Star	#2 - [28, 265]
Optimal Aperture	19.09
Optimal Annulus	39.42
Transit Duration (day)	0.016 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.05 vs 0.139 (deviation 0.23σ)
Duration fit vs published	0.016 d vs 0.0687 d (deviation 2.1σ)
Mid-time O–C	30.4 min
Depth SNR	1.9
Residual scatter	35.5 %

Light curve

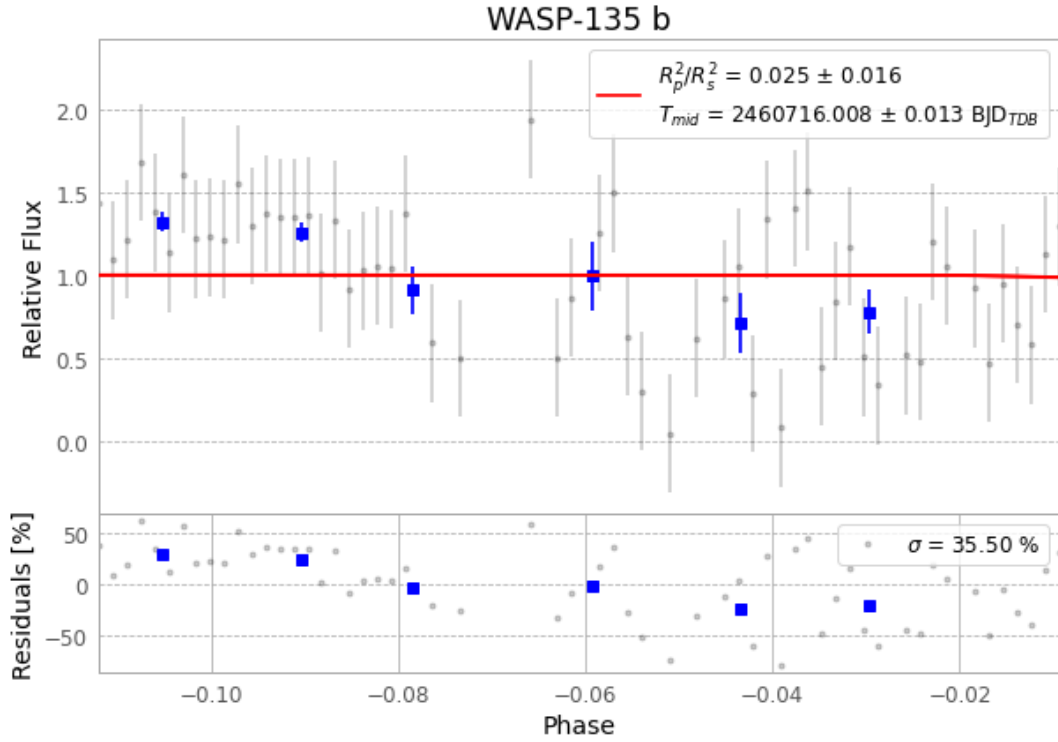


Figure 1 — FinalLightCurve_WASP-135 b_2025-02-09.png (SHA-256 2987f63394e5386c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 405], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d22329e7e965ce7e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit c71eff8

Data provenance

70 FITS frames (46 MB), WASP-135250209082718.FITS → WASP-135250209115417.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250209.log (591cae4d4385...), FinalLightCurve_WASP-135 b_2025-02-09.png (2987f63394e5...), FinalLightCurve_WASP-135 b_2025-02-09.csv (0c4f373f89ae...)

Compute resources

Wall time 320.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 15:04Z.